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Saveetha Institute of Medical And Technical Sciences
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Human resource, payroll , leave and performance management platform

CSA0918 – Programming in Java

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Agenda

- Abstract
- Introduction & Project Overview
- Problem Statement & Gaps in Existing Systems
- Objectives & Proposed Solution System Architecture
- Module Breakdown (HR, Payroll, Leave, Performance)
- System Integration & Flow (Java Backend Architecture)
- Expected Results & Evaluation Plan



Abstract

- **Background:** Organizations frequently struggle with fragmented administrative systems where HR, payroll processing, leave tracking, and employee performance evaluations operate in silos.
- **Core Idea:** A unified, secure, web-based platform built on Java technology to automate workforce management workflows end-to-end.
- **Key Features:** Automated salary slip generation, real-time leave balance calculations, KPI-based performance tracking, and role-based access control (RBAC) for Admin, HR, Managers, and Employees.
- **Impact:** Eliminates manual record-keeping errors, streamlines approval chains, and improves data accuracy across department functions.



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Introduction



- Workforce management requires seamless integration between employee databases and financial operations.
- Digital transformation in internal operations enhances administrative efficiency and reduces overhead costs.
- **Why Java?**
- High enterprise reliability, security, and scalability.
- Robust ecosystem for backend operations (Spring Security, Spring Data JPA).
- Seamless database connectivity via JDBC/Hibernate.



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Problem Identification and Analysis

- **Manual Data Handling:** Reliance on spread-sheets leads to duplicate records and human errors during payroll calculation.
- **Delayed Approvals:** Paper-based or unintegrated email workflows slow down leave requests and performance reviews.
- **Lack of Analytics:** Difficulty in correlating leave trends or performance scores directly with payroll and incentives.
- **Security & Access Risks:** Unsecured data channels present confidentiality risks regarding salary structures and personal data.





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Objectives



- To design and build a secure, web-based platform using Java architecture that unifies core HR, leave management, payroll, and performance tracking into a single system.
- To centralize employee data management by creating a secure relational database for profiles, department details, and organizational roles.
- To automate leave request and approval workflows with real-time balance calculations, automated deductions, and manager notifications.
- To calculate gross and net salaries automatically in Java by integrating basic pay, tax deductions, allowances, and Loss of Pay (LOP) derived from leave data.
- To generate dynamic monthly PDF pay-slips for employees automatically using Java PDF generation libraries.

Solution Design

- **Proposed Architecture:** Model-View-Controller (MVC) Pattern.
- **Backend:** Java (Spring Boot REST APIs / Servlets) for enterprise business logic.
- **Database Layer:** MySQL/PostgreSQL integrated via Spring Data JPA / Hibernate.
- **Frontend:** HTML5, CSS3, JavaScript, Bootstrap (or Thymeleaf/React integration).
- **Security Layer:** Role-Based Access Control (RBAC) with BCrypt password encryption to isolate HR, Manager, and Employee privileges.



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Module 1 : HR & Leave Management

HR Core Module:

- Centralized employee profile management (Onboarding, Personal Info, Designation, Department).
- Unique Employee ID generation and dynamic status tracking (Active, Resigned, On-Leave).

Leave Management System:

- Request submission portal for employees (Paid Leave, Sick Leave, Casual Leave).
- Dynamic leave quota tracking and automatic deductions upon HR/Manager approval.
- Real-time leave calendar and notification triggers.



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Module 2 :Payroll & Performance Tracking

Payroll Processing Engine:

- Automated gross and net salary calculation based on basic pay, allowances, tax deductions, and unpaid leave days.
- PDF pay-slip generation via Java PDF libraries (e.g., iText or JasperReports).

Performance Evaluation Module:

- Quarterly/Annual KPI ratings, Manager feedback submissions, and self-appraisal portals.
- Rating-based incentive mapping directly into the payroll engine.





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Integration of Modules

Inter-Module Connectivity:

- Leave module feeds into the Payroll engine to calculate Unpaid Leaves / Loss of Pay (LOP).
- Performance ratings feed into annual appraisal/bonus calculations within the Payroll engine.

Database Relational Design:

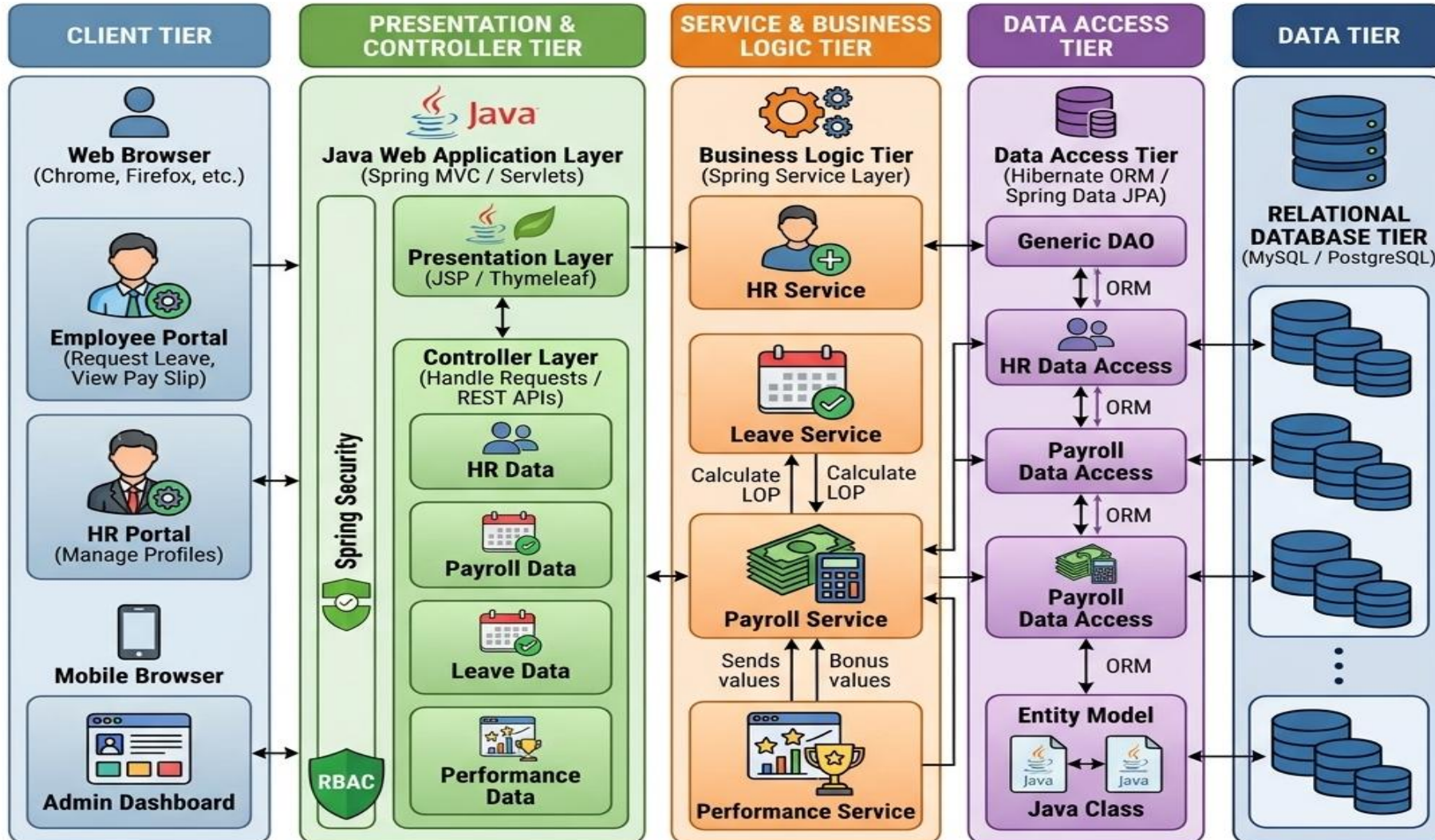
- Centralized relational tables linked via foreign key constraints (employee_id, leave_id, payroll_id, review_id).



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Architecture Diagram



Result and Discussion

- **Automated Calculations:** Zero manual error in monthly payroll processing and leave balance adjustments.
- **Faster Request Processing:** Instant routing of leave and appraisal requests to respective managers.
- **Enhanced Data Security:** Strict data isolation so employees can only access their own pay-slips and leave records.
- **Centralized Reporting:** HR dashboard providing real-time headcount, attendance summaries, and performance distributions.



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Conclusion

- Successfully conceptualized a unified, secure Java-based platform to automate core HR, leave, payroll, and performance operations.
- Developed a robust 3-tier MVC software architecture integrated with Role-Based Access Control (RBAC) to ensure data confidentiality and integrity.
- Designed dynamic inter-module logic to eliminate manual data entry, human calculation errors, and administrative delays.
- Established a solid architectural baseline for Phase 2 development, focusing on database schema deployment, REST API integration, and user interface implementation.

*Thank
you!*